ERCOT Market Learning & Modeling Guide

A systems-first roadmap for building a trading model

Study Plan

2026-02-01

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0.1 0. Prime directive (unchanged, still sacred)

ERCOT prices are **dual variables** of a constrained optimization problem run every 5 minutes under uncertainty.

Everything else is detail.

If your model learns:

- which constraints are likely to bind,
- where they bind (space),
- when they bind (time), and
- how violently they bind (dual magnitude),

then instrument choice (virtuals, PTPs, batteries, AS, spreads) becomes mechanical.

This guide exists to help you model *that*.

0.2 1. ERCOT's role (precision upgrade)

ERCOT operates:

- a **day-ahead financial coordination mechanism**, and
- a **real-time physical control system**,

by repeatedly solving **security-constrained, co-optimized dispatch problems** under uncertainty.

ERCOT is not a marketplace. It is an algorithmic allocator of scarcity.

0.3 2. Market timeline (interpretation tightened)

Your model must reason across **linked inference layers**, not "markets."

0.3.1 2.1 Day-Ahead Market (DAM)

- Clears once per day
- Hourly granularity
- Financial commitments only
- Co-optimizes:
  - Energy

  - Ancillary Services
  - Transmission feasibility

Interpretation

DAM is **not a forecast**. DAM is a **risk-weighted prior** over tomorrow's constraints.

Offers embed:

- startup risk,
- availability risk,
- congestion tail risk,
- fuel and ramp uncertainty.

Persistent DA–RT spreads often reflect *insurance premia*, not error.

0.3.2 2.2 Reliability Unit Commitment (RUC)

- DRUC / HRUC
- Operator-driven, not profit-driven
- Commits capacity for feasibility, not efficiency

Interpretation

RUC is **constraint pre-activation**.

It reveals:

- operator distrust of DAM feasibility,
- looming ramp or reserve stress,
- spatial inadequacy that prices have not yet expressed.

RUC is a *signal generator*, not a trading venue.

0.3.3 2.3 Real-Time Market (RTM)

- Clears every 5 minutes
- Physical dispatch
- Solves SCED with:
  - energy balance,
  - congestion,
  - ramping,
  - reserve and flexibility constraints

Interpretation

RT is not volatility. RT is **constraint revelation**.

RT prices are where beliefs meet physics.

0.4 3. The core optimization (make this explicit)

At each interval, ERCOT solves:

> Minimize system cost subject to feasibility and security constraints

0.4.1 3.1 Prices are duals, not opinions

- LMP = dual of nodal balance
- Congestion = dual of transmission constraints
- Scarcity = dual of reserve / infeasibility constraints

Prices are **shadow prices of marginal infeasibility**.

This is the conceptual hinge of the entire model.

0.5 4. Locational Marginal Pricing (formalized)

For node n at time t:

$$\text{LMP}_{n,t} = \lambda_t + \sum_k \mu_{k,t} \cdot \text{PTDF}_{k,n} + \text{losses}$$

Where:

- λ_t: system-wide marginal energy feasibility
- $\mu_{k,t}$: shadow price of constraint k

Modeling implication

Do **not** forecast LMPs directly.

Forecast:

- which μ_k activate,
- their sign,
- their order of magnitude,
- their spatial footprint.

Prices fall out automatically.

0.6 5. Congestion, PTPs, CRRs (spatial optionality)

0.6.1 5.1 Congestion = binding network constraint

When a constraint binds:

- prices separate spatially,
- congestion rent appears,
- dual variables become monetizable.

0.6.2 5.2 Financial interpretation

A PTP Obligation pays:

$$(\mathrm{LMP}_{\mathrm{sink}} - \mathrm{LMP}_{\mathrm{source}}) \times \mathrm{MW}$$

This is a **linear functional of constraint duals**.

0.6.3 5.3 Modeling takeaway

Congestion trading is **constraint classification**, not price prediction.

CRRs/PTPs are basis vectors in dual-space.

0.7 6. Ancillary Services (RTC-era framing)

Ancillary Services are **constraints purchased in advance**.

Under Real-Time Co-Optimization:

- Scarcity is priced *inside* the optimization
- AS demand curves replace ex-post adders
- AS duals bleed directly into LMPs

0.7.1 6.1 Key insight

Energy and reserves are **no longer separable markets**.

They are competing uses of the same feasibility region.

0.7.2 6.2 Modeling implication

You must forecast **joint energy–reserve feasibility**, not:

- “energy prices” plus
- “scarcity adders.”

0.8 7. Renewables (what actually changes)

Renewables introduce:

- stochastic injections,
- forecast error,
- spatial clustering,
- ramp stress,
- inertia loss.

Consequences:

- mean prices ↓
- variance ↑
- tail risk ↑
- constraint activation frequency ↑

Renewables do not break markets. They **excite dual variables**.

0.9 8. Storage & batteries (temporal duality)

A battery is:

- a temporal arbitrage device
- with hard intertemporal constraints (SoC, duration)

0.9.1 8.1 Value sources

- Energy time spreads
- Congestion exposure (location)
- Ancillary flexibility
- Scarcity response

0.9.2 8.2 Structural mirror

- Transmission → spatial constraint
- SoC → temporal constraint

Same math. Different axis.

0.10 9. Participant-side optimization (clean abstraction)

Market participants solve a **two-stage stochastic control problem**:

- Stage 1: DAM commitments (risk-weighted)
- Stage 2: RT realization and correction

Offers are:

- not forecasts,
- but **risk-adjusted feasibility declarations**.

0.11 10. Risk management (non-optional)

ERCOT markets exhibit:

- fat tails,
- regime shifts,
- correlated failures,
- binding credit constraints.

Expected PnL is insufficient.

CVaR framing is essential:

- weight tail outcomes,
- enforce survival constraints,
- size positions under stress, not comfort.

A good model maximizes **survivable profit**, not mean profit.

0.12 11. Modern AS offering under uncertainty

Flexible resources must satisfy:

- duration constraints,
- probabilistic availability,
- performance penalties.

This turns bidding into:

> **quantile forecasting under joint chance constraints**

Your model must reason about *availability distributions*, not point capability.

0.13 12. Minimum viable ERCOT trading architecture (explicit)

0.13.1 12.1 World model (belief state)

Latent state S_t includes:

- load deviation regime,
- renewable error regime,
- outage stress,
- transmission availability,
- reserve tightness.

This is low-dimensional regime inference, not brute force ML.

0.13.2 12.2 Constraint forecaster (the core)

Predict:

- probability a constraint binds,
- expected dual magnitude bucket,
- spatial footprint (PTDF exposure),
- temporal persistence.

This is classification + calibration, not regression.

0.13.3 12.3 Decision layer

Choose:

- DA vs RT exposure,
- spatial spreads (PTPs, virtuals),
- temporal spreads (storage),
- AS participation,
- position size under CVaR.

Once duals are forecast, this layer is algebra.

0.13.4 12.4 Learning loop

1. Observe outcomes
2. Update belief state
3. Update constraint activation models
4. Re-price risk and resize exposure

This is belief $\rightarrow$ constraint $\rightarrow$ dual $\rightarrow$ instrument $\rightarrow$ belief.

0.14 13. Naming the algorithm (now precise)

What you are building is:

> **A belief-driven constraint-activation engine with dual-space trading**

In plainer terms:

1. Infer uncertainty regimes
2. Forecast constraint activation
3. Estimate shadow prices
4. Trade instruments that isolate those shadows
5. Control downside explicitly

This is not price prediction. This is **market microstructure-aware control**.

0.15 14. Why this order of learning is correct

Most people:

- learn prices,
- then back-explain constraints.

You are:

- learning constraints,
- then letting prices emerge.

That means when you encounter:

- stochastic control,
- derivatives,
- real options,
- power economics,

you won't memorize formulas. You'll recognize structures.

0.16 15. Recommended next steps (tightened)

The highest-leverage next moves:

1. Write a **constraint taxonomy** (one page)
2. Pick **one instrument class** (virtuals *or* batteries)
3. Formalize the **belief update loop** (even pseudocode)
4. Add CVaR sizing
5. Only then optimize PnL

No rush. This compounds.

0.17 Final thought (unchanged, still true)

You are not trying to beat ERCOT.

You are trying to listen to it earlier than others.

Markets speak in prices. Prices speak in duals. Duals speak in constraints.

You are learning the language fluently.

Below is **one-page ERCOT constraint taxonomy**, written the way a model would want to see the world: not as market products, but as **families of constraints**, their observables, their duals, and the instruments that expose them.

This is deliberately compact and structural. You should be able to pin this above your desk or drop it into a Quarto appendix unchanged.

1 ERCOT Constraint Taxonomy

(Model-centric, RTC-era)

All ERCOT prices emerge from a single repeated optimization run by ERCOT. This taxonomy classifies **what can bind**, **why it binds**, and **how its shadow price is monetized**.

1.1 1. Energy Balance Constraints (System-wide)

What it is The requirement that total generation + imports = total load + losses.

Why it binds

- Aggregate supply scarcity
- Fuel outages
- Extreme load
- Renewable underperformance

Primary observables

- System load forecast error
- Available online capacity
- Net load ramps
- Reserve depletion trajectory

Dual variable

- System lambda (λ_t)

Price manifestation

- System-wide LMP component
- Broad price elevation across nodes

Tradable exposure

- Virtual energy positions (DA–RT)
- System-wide batteries

- Energy-only generation

Modeling note This is *not* "energy price forecasting." This is forecasting **system infeasibility pressure**.

1.2 2. Transmission / Network Constraints (Spatial)

What it is Limits on power flows across monitored transmission elements.

Why it binds

- Line outages
- High regional injections (wind/solar clustering)
- Load pockets
- Topology changes

Primary observables

- Outage reports
- PTDF maps
- Regional generation/load imbalance
- Historical constraint activation patterns

Dual variable

- Constraint shadow price ($\mu_{k,t}$)

Price manifestation

- Spatial LMP separation
- Persistent nodal spreads

Tradable exposure

- PTP Obligations
- CRRs
- Location-specific virtuals
- Batteries with locational advantage

Modeling note Congestion trading = predicting **which constraint index (k)** lights up, not prices.

1.3 3. Ramping Constraints (Temporal, short-horizon)

What it is Limits on how fast resources can change output.

Why it binds

- Sunrise / sunset ramps
- Wind drop-offs
- Unit trips
- Net load volatility

Primary observables

- Net load ramp forecasts
- Generator ramp rates
- Intra-hour renewable volatility

Dual variable

- Implicit ramp shadow prices (often folded into energy/reserve duals)

Price manifestation

- Fast price spikes
- Short-lived scarcity
- Extreme 5-minute volatility

Tradable exposure

- Fast-responding batteries
- Regulation services
- Short-duration virtuals

Modeling note Ramping constraints explain **violence**, not persistence.

1.4 4. Reserve Adequacy Constraints (Flexibility)

What it is Requirements to hold sufficient upward/downward flexibility to manage uncertainty.

Why it binds

- Forecast uncertainty
- Unit failures
- High renewable penetration
- Tight operating margins

Primary observables

- Available reserve levels
- Reserve procurement outcomes
- AS Demand Curve position
- Deployment frequency

Dual variable

- Reserve / AS shadow prices (RTC-embedded)

Price manifestation

- Ancillary Service prices
- Scarcity embedded directly into LMPs (RTC)

Tradable exposure

- AS offers (Reg, RRS, ECRS, Non-Spin)
- Batteries
- Flexible load

Modeling note Under RTC, reserve scarcity is **inside the optimization**, not an add-on.

1.5 5. Unit Commitment & Minimum Output Constraints (Operational)

What it is Binary and semi-binary constraints on whether units can be online and at what minimum levels.

Why it binds

- Startup times
- Minimum run times
- Minimum stable generation
- Fuel limitations

Primary observables

- RUC commitments
- Unit status flags
- Startup cost signals

Dual variable

- Reflected indirectly through prices and uplift logic

Price manifestation

- DAM vs RT bias
- Price floors or suppressed volatility

Tradable exposure

- DAM–RT spreads
- Longer-horizon virtuals

Modeling note These constraints explain **system stickiness**, not spikes.

1.6 6. Energy-Limited Resource Constraints (Intertemporal)

What it is State-of-charge and duration limits for batteries and energy-limited units.

Why it binds

- Sustained high prices
- Consecutive scarcity intervals
- Poor charging opportunities

Primary observables

- SOC telemetry
- Duration qualification
- Historical dispatch sequences

Dual variable

- Intertemporal shadow prices (implicit)

Price manifestation

- Scarcity persistence
- Depletion-driven price cliffs

Tradable exposure

- Batteries
- Storage-backed AS positions

Modeling note This is the **time-axis analog of congestion**.

1.7 7. Credit & Risk Constraints (Participant-side, binding before market)

What it is Limits on position size imposed by collateral and credit rules.

Why it binds

- Volatility
- Concentrated positions
- Adverse scenarios

Primary observables

- Exposure calculations
- Stress scenarios
- Margin requirements

Dual variable

- Not priced by ERCOT, but binds trader behavior

Price manifestation

- Forced de-risking
- Liquidity gaps

Tradable exposure

- Position sizing only

Modeling note These constraints determine **who survives**.

1.8 Structural Summary (the mental compression)

- **Energy constraints** → system lambda
- **Transmission constraints** → spatial duals
- **Ramping constraints** → volatility
- **Reserve constraints** → scarcity pricing
- **Commitment constraints** → bias & inertia
- **Intertemporal constraints** → persistence
- **Credit constraints** → survival

Every ERCOT trade is a **projection of one or more of these duals**.

1.8.1 Final modeling instruction

Do not ask:

> "What will the price be?"

Ask:

> "Which constraint family is approaching its boundary, where, and for how long?"

The answer determines the instrument automatically.

Below is the **constraint-to-data map**, written so a model builder can go straight from *conceptual constraint → ERCOT feeds → features.* This is the missing wiring diagram between theory and code.

2 ERCOT Constraint → Data Feed Map

(What to ingest, why it matters, how the model uses it)

All feeds below are produced directly or indirectly by ERCOT. The organizing principle is simple:

> **If a constraint can bind, there must exist telemetry or disclosure that hints it is approaching the boundary.**

2.1 1. Energy Balance Constraints (System Lambda)

Constraint family System-wide supply = demand + losses

2.1.1 Primary ERCOT feeds

- **System Load Forecast** (STLF, MTLF)
- **Actual System Load**
- **Resource Status & Availability**
- **Outage Scheduler (planned + forced)**
- **Real-Time System Lambda / System LMP**

2.1.2 Model features

- Load forecast error (actual − forecast)
- Available capacity margin
- Forced outage rate
- Net load level and slope
- Frequency of scarcity intervals

2.1.3 What the model infers

- Probability of system infeasibility
- Direction and magnitude bucket of λ_t

2.2 2. Transmission / Network Constraints (Congestion)

Constraint family Thermal / security limits on monitored elements

2.2.1 Primary ERCOT feeds

- **Shift Factor (PTDF) Extracts**
- **Constraint & Limit (C&L) Reports**
- **Binding Constraint Reports**
- **Outage Scheduler (transmission)**
- **Nodal LMPs**

2.2.2 Model features

- Historical constraint activation frequency
- PTDF exposure by node/zone
- Topology change indicators
- Correlation between injections and congestion

2.2.3 What the model infers

- Which constraint index (k) is likely to bind
- Expected sign and size of μ_k
- Spatial footprint of congestion rent

2.3 3. Ramping Constraints (Short-Horizon Feasibility)

Constraint family Limits on how fast the system can rebalance

2.3.1 Primary ERCOT feeds

- **Net Load Forecasts (intra-hour)**
- **Wind & Solar Generation Forecasts**
- **Actual Renewable Output**
- **5-Minute SCED Dispatch**
- **Regulation Deployment Data**

2.3.2 Model features

- Net load ramp rate (MW/min)
- Renewable forecast error variance
- Generator ramp capability vs required ramps
- Frequency of regulation saturation

2.3.3 What the model infers

- Probability of ramp scarcity
- Likelihood of short-lived price spikes
- Value of fast response assets

2.4 4. Reserve Adequacy Constraints (Flexibility / Scarcity)

Constraint family Holding sufficient upward/downward reserves

2.4.1 Primary ERCOT feeds

- **Ancillary Service Demand Curves (ASDCs)**
- **Ancillary Service Awards & Prices**
- **Available Reserve Levels**
- **AS Deployment Events**
- **SCED Reserve Co-optimization Outputs**

2.4.2 Model features

- Reserve margin vs ASDC knee
- Frequency of reserve deployments
- AS price convexity
- Correlation between reserve tightness and LMPs

2.4.3 What the model infers

- Embedded scarcity pressure
- Joint energy–reserve stress
- When AS duals will dominate pricing

2.5 5. Unit Commitment & Minimum Output Constraints

Constraint family Binary and minimum-run operational limits

2.5.1 Primary ERCOT feeds

- **RUC Results (DRUC / HRUC)**
- **Committed Unit Lists**
- **Startup & Shutdown Indicators**
- **DAM Awards vs RT Dispatch**

2.5.2 Model features

- RUC override frequency
- Commitment bias (DAM vs RT)
- Unit inflexibility measures
- Offline capacity forced online

2.5.3 What the model infers

- Structural rigidity in supply
- Bias toward price suppression or elevation
- Multi-hour persistence risk

2.6 6. Energy-Limited Resource Constraints (Intertemporal)

Constraint family State-of-charge and duration limits

2.6.1 Primary ERCOT feeds

- **Energy Storage Resource (ESR) Telemetry**
  - SOC
  - Max/Min SOC
  - Charge/Discharge Limits
- **ESR Dispatch & AS Awards**
- **Duration Qualification Data**

2.6.2 Model features

- SOC trajectory vs price
- Depletion probability
- Consecutive high-dispatch intervals
- Charging opportunity windows

2.6.3 What the model infers

- Risk of storage exhaustion
- Persistence of scarcity after initial spike
- When temporal duals dominate

2.7 7. Credit & Risk Constraints (Participant-Side)

Constraint family Capital and collateral limits (bind *before* market clearing)

2.7.1 Primary ERCOT feeds

- **Credit Exposure Reports**
- **Historical Price Distributions**
- **Stress Scenario Definitions**
- **Settlement & Margin Data**

2.7.2 Model features

- Tail price scenarios
- Exposure concentration
- Scenario-weighted losses
- CVaR by instrument

2.7.3 What the model infers

- Maximum survivable position size
- When liquidity will vanish
- When "right trades" become untradable

2.8 Compression: one mental model

Constraint Type	Signal Source	Dual Revealed As	Primary Instrument
Energy	Load / capacity	System λ	Virtual energy
Transmission	PTDF + outages	μ_k	PTPs / CRRs
Ramping	Net load slope	Volatility	Fast assets
Reserves	ASDC position	Scarcity	AS / batteries
Commitment	RUC actions	Bias	DA–RT spreads
Intertemporal	SOC	Persistence	Storage
Credit	Margin stress	Survival	Position size

2.9 Final instruction for implementation

For each constraint family, your pipeline should answer **one binary question** every interval:

> *Is this constraint approaching its feasible boundary?*

Everything else — prices, spreads, PnL — is downstream.